Vehicle Orientation Estimation Using Conventional Image Processing

Het Patel
Enrollment No.: AU2440221
B.Tech. CSE
Ahmedabad University

Shlok Shah
Enrollment No.: AU2440178
B.Tech. CSE
Ahmedabad University

Malay Panchal
Enrollment No.: AU2440050
B.Tech. CSE
Ahmedabad University

Dev Prajapati
Enrollment No.: AU2440022
B.Tech. CSE
Ahmedabad University

Group 14, Digital Image Processing Project

***Abstract*—This paper studies the problem of estimating vehicle orientation by using conventional image processing based on top-down aerial images. Experiments were done using the DRASHTI-HaOBB dataset, which provides four corner vehicle annotations, vehicle class and ground-truth heading. Each vehicle is cropped and converted to grayscale, Sobel gradients are calculated, and the dominant orientation is found from a magnitude-weighted histogram of gradient directions and compared with ground-truth headings. On 20 cropped vehicle images, the mean absolute error (MAE) is 33.9°, with minimum and maximum errors of 1.32° and 67.38°. The Sobel method gives close estimates in some cases but large errors overall, which shows the limits of dominant image gradients as an estimate for vehicle orientation and motivates to rather use a structure-oriented method such as the Radon transform used in the literature.**

***Index Terms*—Aerial images, conventional image processing, gradient magnitude, Radon transform, Sobel operator, vehicle orientation estimation.**

I. Introduction

Traffic surveillance and urban planning based on drone images needs the heading of each vehicle, for example to detect wrong-way driving or estimate movement, especially in unstructured traffic where vehicles do not follow lanes. The DRASHTI-HaOBB dataset [1] provides 27,577 nadir-view (straight-down) drone images (45–127 m altitude, 14 vehicle classes), with an oriented bounding box and a heading angle for each vehicle.

Liu et al. [2] note that low-level features like edges, intensities and shadows might not be enough for complex aerial backgrounds, review gradient-direction-histogram and Canny/Hough line-angle approaches and apply the Radon transform for finding a vehicle region's principal orientation. In our work, we use only this orientation concept and start testing with Sobel method on 20 vehicle crops, measuring how accurate its results are compared to ground-truth orientations.

II. Methodology

A. Dataset and Sample Selection

Each annotation has four corner points, a class, a difficulty flag and a heading angle φ in range of [0°, 360°); the first line of each file gives the flight height. We have drawn the boxes and heading arrows and cropped each vehicle with 8 pixels of padding (Fig. 1).



Fig. 1. Boxes and heading arrows drawn from the annotations (left) and four cropped vehicles (right).

B. Ground-Truth Angle Conversion

For evaluation, each annotated heading φ is converted to an unsigned orientation θ_{gt} between 0° and 90°, which treats orientations along the same undirected axis as equivalent: the heading is reduced modulo 180° and, if the result exceeds 90°, subtracted from 180°,

$$\theta_{\text{gt}} = \begin{cases} \varphi \bmod 180°, & \text{if } \varphi \bmod 180° \le 90° \\ 180° - (\varphi \bmod 180°), & \text{otherwise} \end{cases} \quad (1)$$

C. Sobel Baseline

Each vehicle crop is converted to a grayscale image I and filtered with the Sobel kernels

$$K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad K_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad (2)$$

to obtain the horizontal and vertical gradient components $G_x = K_x * I$ and $G_y = K_y * I$ (Fig. 2), where $*$ denotes 2-D filtering. The gradient magnitude and direction are

$$G = \sqrt{G_x^2 + G_y^2} \quad (3)$$

$$\theta = \operatorname{atan2}(G_y, G_x) \quad (4)$$

A magnitude weighted histogram with 180 bins over -90° to 90° counts the gradient magnitude at each pixel's direction, and the direction of the maximum bin is taken as the dominant gradient direction θ_{est}. Sobel finds only local gradients, so this approach assumes that the dominant direction represents orientation of vehicle.

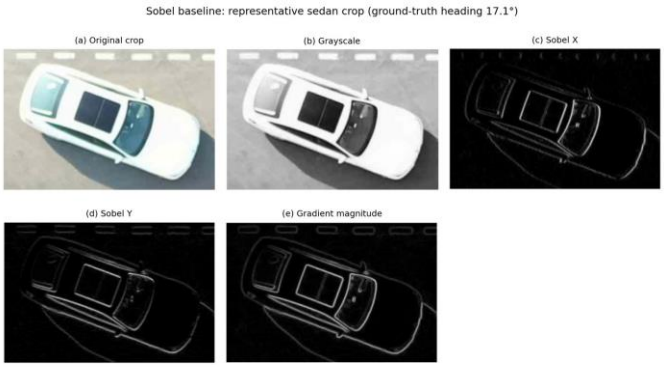


Fig. 2. Sobel steps for the sedan crop.

D. Error Calculation and Evaluation

The absolute error of vehicle i and the mean absolute error over the $N = 20$ evaluated crops are

$$E_i = \left|\theta_{\text{est}} - \theta_{\text{gt}}\right| \quad (5)$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} E_i \quad (6)$$

The minimum and maximum errors over the sample are also reported.

III. Results

For the sedan cropped image (Fig. 2), the ground truth is 17.1° and the estimate is approximately 11°, an absolute error of about 6.1° (v01 in Table I). Over all 20 crops (Table I I,), the MAE is 33.9°, the minimum error is 1.32° (v03) and the maximum is 67.38° (v05).

TABLE I
SOBEL ORIENTATION ESTIMATES FOR 20 VEHICLE CROPS

Crop ID	Class	Ground truth (°)	Axis angle (°)	Sobel estimate (°)	Error (°)
v01	Sedan	17.10	17.10	11.0	6.10
v02	SUV	181.38	1.38	63.0	61.62
v03	Motorcycle	1.68	1.68	3.0	1.32
v04	Motorcycle	36.32	36.32	5.0	31.32
v05	Motorcycle	183.62	3.62	71.0	67.38
v06	Sedan	270.00	90.00	23.0	67.00
v07	SUV	181.88	1.88	51.0	49.12
v08	Motorcycle	43.68	43.68	21.0	22.68
v09	Pickup	178.04	1.96	48.0	46.04
v10	Motorcycle	270.00	90.00	79.0	11.00
v11	Motorcycle	270.00	90.00	71.0	19.00
v12	Motorcycle	270.00	90.00	36.0	54.00
v13	SUV	250.00	70.00	33.0	37.00
v14	Van	0.00	0.00	29.0	29.00
v15	Auto-rickshaw	79.90	79.90	27.0	52.90
v16	Van	101.40	78.60	51.0	27.60
v17	Motorcycle	77.90	77.90	73.0	4.90
v18	Auto-rickshaw	294.30	65.70	50.0	15.70
v19	Auto-rickshaw	0.00	0.00	45.0	45.00
v20	Tanker	177.40	2.60	32.0	29.40

\xis angle is the heading folded to 0°–90° by (1).

TABLE II
SOBEL BASELINE ON 20 VEHICLE CROPS.

Metric	Value
Mean absolute error	33.9°
Minimum error	1.32°
Maximum error	67.38°

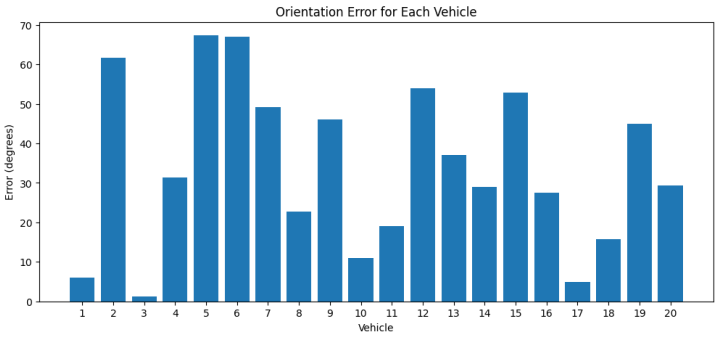


Fig. 3. Orientation error for each vehicle.

IV. Discussion

The sedan vehicle example (6.1° error) does not represent the sample (33.9° MAE). The 1.32° minimum shows that some close estimates by Sobel operator is possible, but the 67.38° maximum shows that large deviations could also occur, so the dominant gradient direction is not a consistently accurate representation of vehicle orientation.

A possible explanation for this inaccuracy is that the Sobel operator responds to local intensity changes rather than to the vehicle's longitudinal axis. Vehicle boundaries, roof edges, windows, shadows and background structures all contribute, so the strongest accumulated direction may differ from the true front-to-back direction; these are possible, not individually verified, causes. The Radon transform is a next step for extracting dominant directional structure, but it has not been evaluated here.

V. Conclusion

In this paper, we have researched vehicle orientation estimation on the DRASHTI-HaOBB dataset using Sobel gradients and a magnitude weighted direction histogram. On 20 vehicle crops the MAE was 33.9° (minimum 1.32°, maximum 67.38°), so the method gives close estimates in some cases but is not consistently accurate. In future work, we will investigate alternative image processing methods such as the Radon transform and evaluate them on more crops.

References

[1] Y. M. Bhavsar, M. S. Zaveri, M. S. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone nadir-view annotated images of vehicles detection dataset for India with heading-angle oriented bounding box (DRASHTI-HaOBB)," *IEEE Data Descriptions*, vol. 3, pp. 192–202, Mar. 2026, doi: 10.1109/IEEEDATA.2026.3670752.

[2] C. Liu, Y. Ding, M. Zhu, J. Xiu, M. Li, and Q. Li, "Vehicle detection in aerial images using a fast oriented region search and the vector of locally aggregated descriptors," *Sensors*, vol. 19, no. 15, p. 3294, Jul. 2019, doi: 10.3390/s19153294.